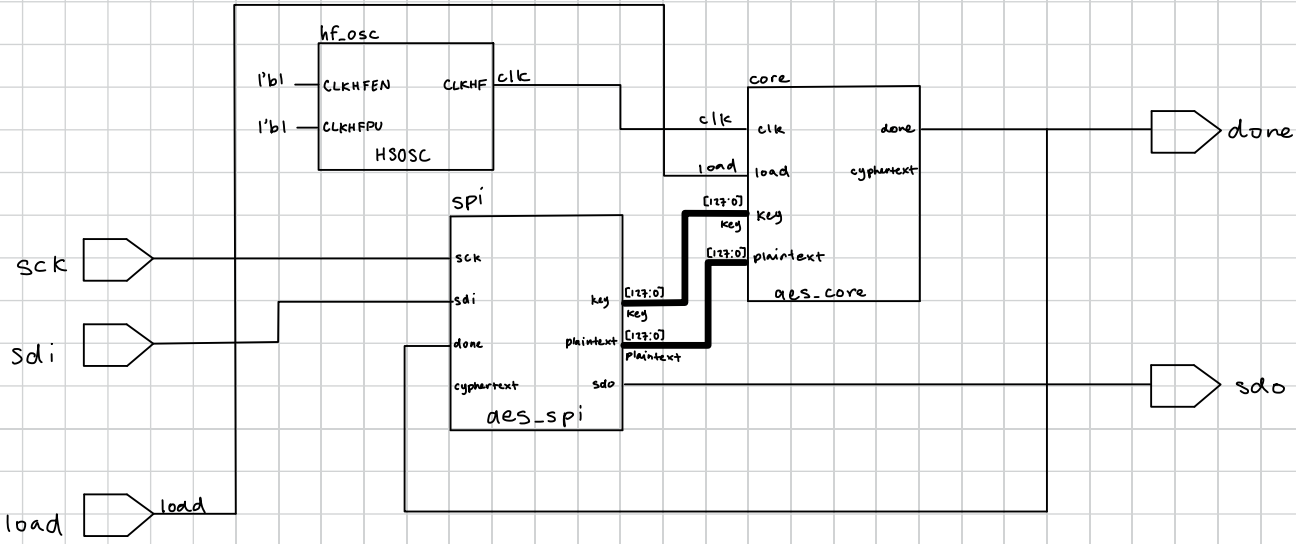
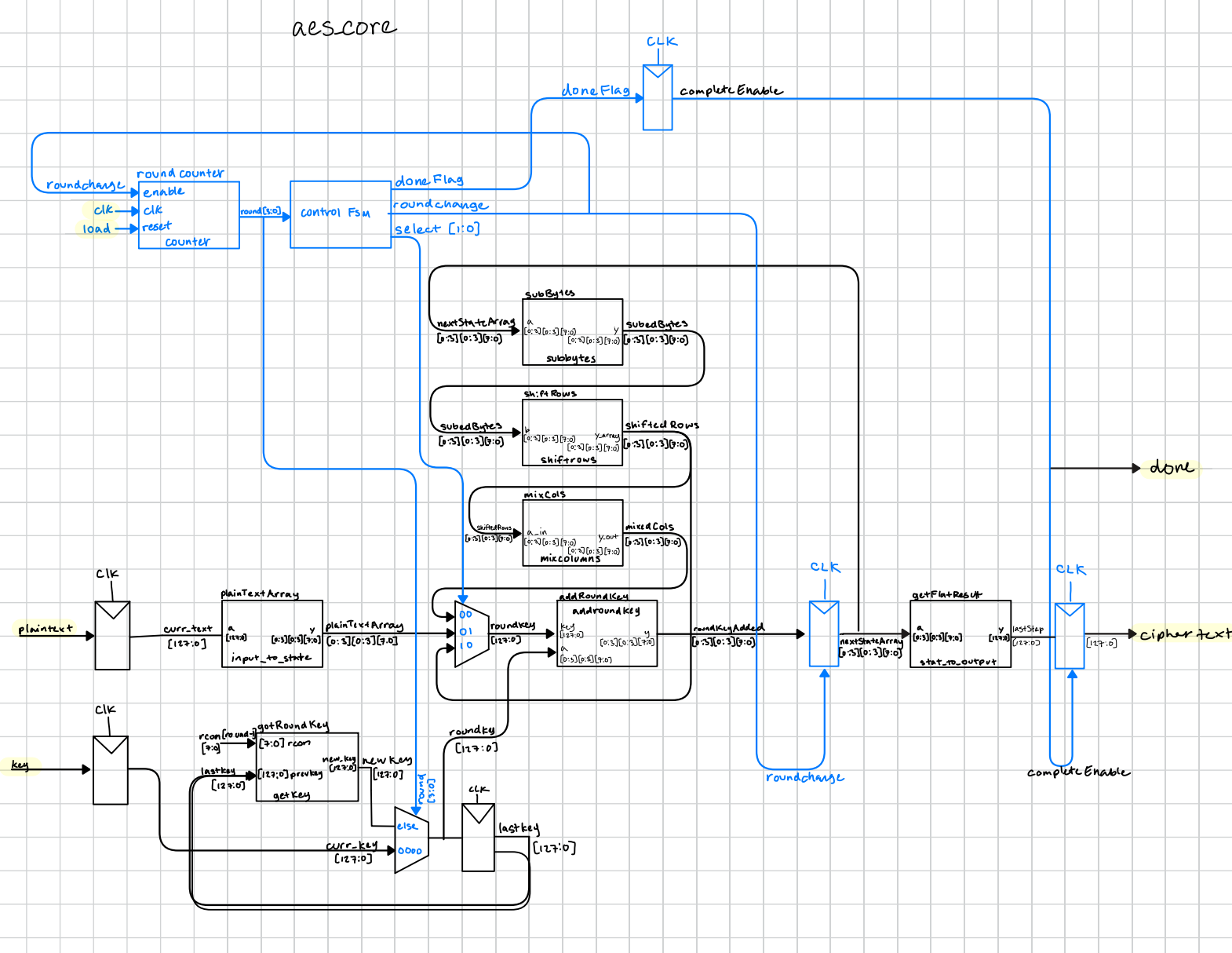
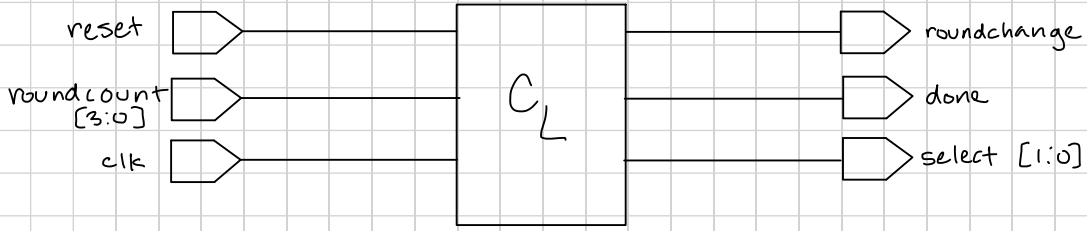
Top Module – AES

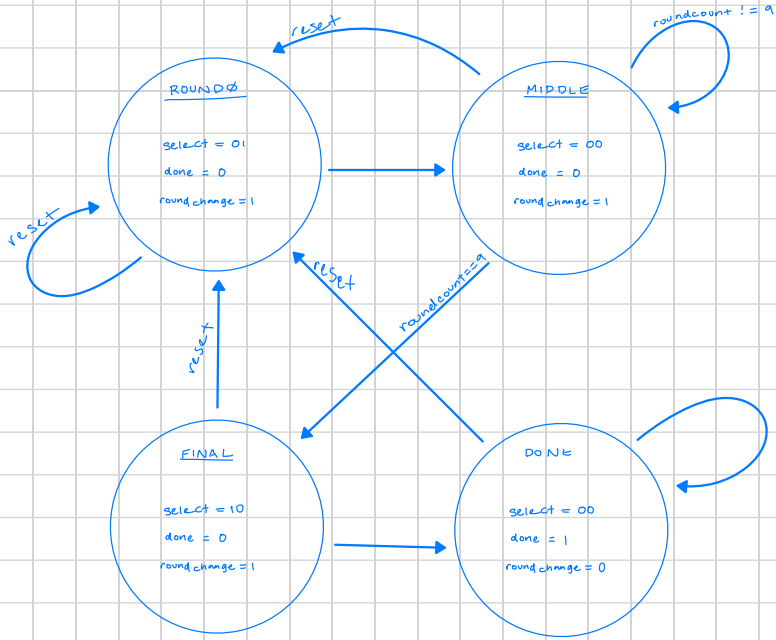



aes_core
CLK
doneFlag
completeEnable
roundchange
round counter
enable
clk
load
reset
counter
round[3:0]
control FSM
doneFlag
roundchange
select [1:0]
subBytes
nextStateArray
[0:3][0:3][7:0]
subbytes
subedBytes
[0:3][0:3][7:0]
shiftRows
subedBytes
shiftrows
shifted Rows
mixCols
shiftedRows
mixcolumns
mixedCols
done
CLK
plainText
plainText Array
curr_text
[127:0]
input_to_state
00
01
10
roundkey
[127:0]
addRoundKey
addroundkey
roundKeyAdded
CLK
nextStateArray
getFinalResult
stat_to_output
lastStep
CLK
cipher text
key
CLK
rcon[round]
[7:0]
gotRoundKey
lastkey
[127:0]
getkey
newkey
[127:0]
round
[3:0]
else
0000
curr_key
[127:0]
CLK
lastkey
[127:0]
roundkey
[127:0]
roundchange
completeEnable

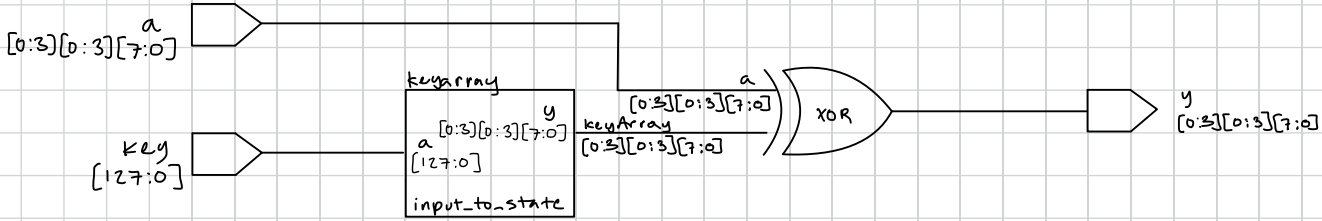
Contrd FSM

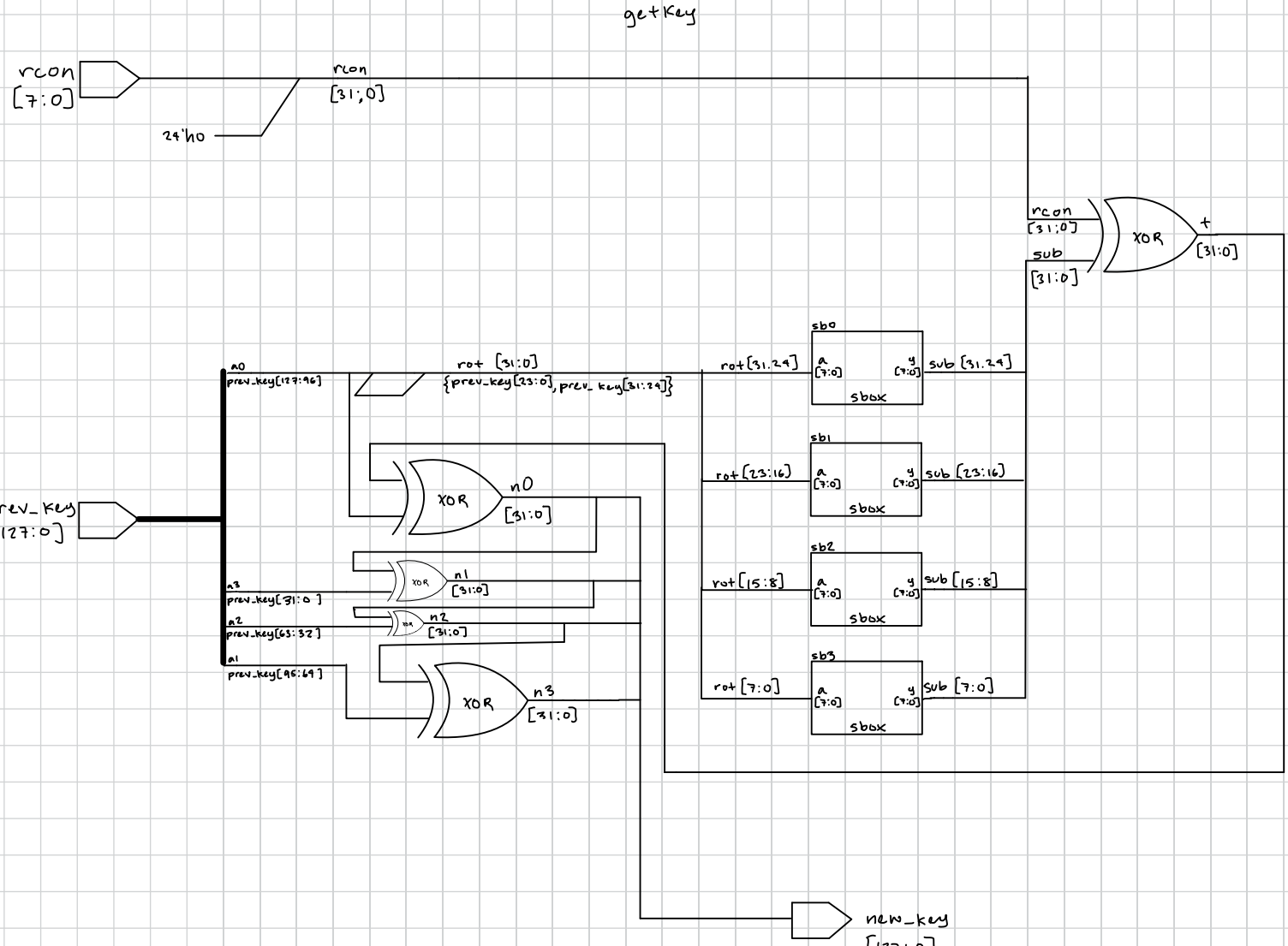


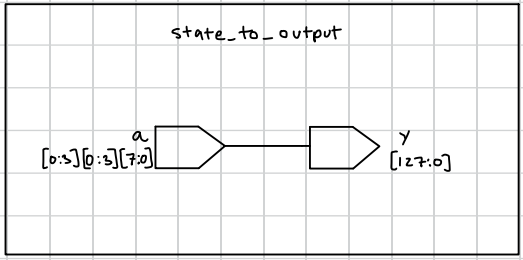
State Transition Diagram:

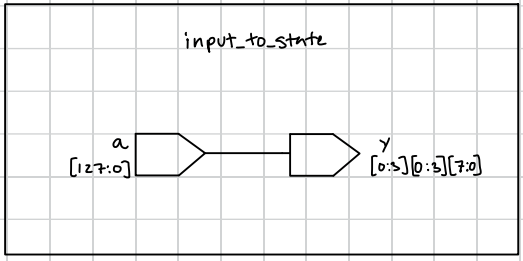


addRoundKey




getKey
rcon
[7:0]
rcon
[31:0]
24'h0
rcon
[31:0]
sub
[31:0]
XOR
t
[31:0]
sb0
sb1
sb2
sb3
a
[7:0]
y
[7:0]
sbox
rot [31:24]
sub [31:24]
rot [23:16]
sub [23:16]
rot [15:8]
sub [15:8]
rot [7:0]
sub [7:0]
a0
prev_key[127:96]
rot [31:0]
{prev_key[23:0], prev_key[31:24]}
prev_key
[127:0]
XOR
n0
[31:0]
a3
prev_key[31:0]
XOR
n1
[31:0]
a2
prev_key[63:32]
XOR
n2
[31:0]
a1
prev_key[95:64]
XOR
n3
[31:0]
new_key
[127:0]
{n0, n1, n2, n3}


state_to_output
a
[0:3][0:3][7:0]
y
[127:0]


input_to_state
a
[127:0]
y
[0:3][0:3][7:0]


shiftrows
b
[0:3][0:3][7:0]
y_array
[0:3][0:3][7:0]

subbytes

a
[0:3][0:3][7:0]
y
[0:3][0:3][7:0]
y00
y01
y02
y03
y10
y11
y12
y13
y20
y21
y22
y23
y30
y31
y32
y33
sbox
a [7:0]
y [7:0]